

Introduction to AI Ethics and Governance

WHY THIS MODULE MATTERS

Almost every organisation is now using AI somewhere, and in most of them nobody has written down where. This module gives you the vocabulary, the distinction between ethics and governance, and an honest picture of why this became urgent.

1.1 What This Course Is

This is a course about the decisions an organisation makes when it allows software to decide something on its behalf — and about who is answerable when that decision turns out badly. It assumes no technical background and no legal training.

Artificial intelligence has stopped being a specialist concern. Systems that rank job applicants, price insurance, flag suspicious transactions, triage patients, moderate content, forecast demand, draft documents and answer customers are now ordinary infrastructure in UK organisations of every size. Most were adopted for straightforward reasons: they are faster, cheaper, and they scale. Very few were adopted with a clear answer to the question of what happens when they are wrong.

That gap is what this course addresses. It is deliberately cross-sector. The examples come from recruitment, education, welfare, policing, aviation, manufacturing and retail, because the failure patterns are remarkably consistent across all of them. A system that learns the wrong lesson from historic data behaves the same way whether it is sorting CVs or sorting insurance claims. The mitigations are consistent too.

The course is practical rather than philosophical. It will not resolve longstanding ethical debates, and it does not need to. What it will give you is a working method: how to identify which AI systems your organisation relies on, how to judge which of them carry real risk, what the UK legal framework actually requires, and what a defensible process looks like when someone asks you to justify a decision.

1.2 A Working Definition of AI

"Artificial intelligence" is used loosely, and the looseness causes real problems in governance because it makes it hard to say what is in scope. For this course, an AI system is any system that uses data to produce outputs — predictions, scores, classifications, recommendations, decisions or generated content — through patterns it has derived from that data rather than through rules a person wrote out explicitly.

That definition matters because it captures things people do not always think of as AI. A spreadsheet model with hand-written rules is not AI under this definition. A scoring tool that a vendor trained on ten years of your industry's data is, even if the interface looks like a spreadsheet. The governance question is not what the technology is called; it is whether anyone can explain why it produced the output it did.

- **Machine learning** — a system that derives its own rules from training data. The majority of AI in commercial use is of this kind.
- **Generative AI** — a system that produces new content, most commonly text or images, from a prompt. Covered in Module 3.
- **Automated decision-making** — a legal term rather than a technical one, referring to a decision made about a person without meaningful human involvement. It carries specific obligations under UK data protection law, covered in Module 5.
- **AI agent** — a system permitted to take actions, not merely make recommendations. The governance implications are different, and greater.

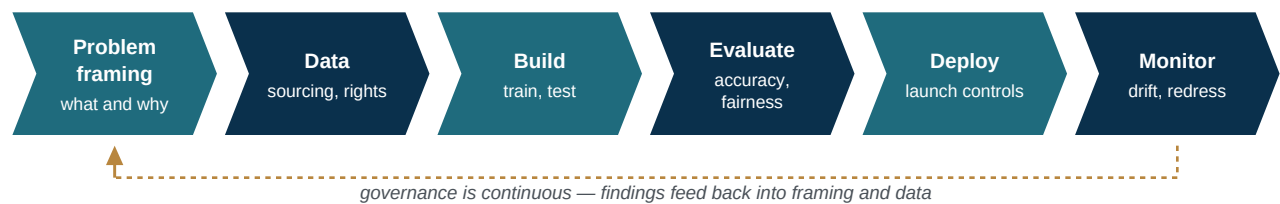


Figure 1.1 — The AI system lifecycle. Governance applies at every stage, not only at deployment.

1.3 Ethics and Governance Are Not the Same Thing

These two words are used almost interchangeably in most organisations, which is part of the reason both are so often neglected. They describe different activities and they fail in different ways.

AI ethics refers to the reasoning used to judge whether a system’s design and behaviour are justifiable given their effects on people. It is the standard-setting activity: what would count as fair here, what would count as adequate transparency, what level of error is acceptable and to whom.

AI governance refers to the structures, records and processes that make an organisation actually meet that standard, consistently, and demonstrate that it did. It is the operational activity: who approved this, what was checked before launch, who monitors it now, what triggers a review, who is accountable when it fails.

Ethics without governance produces a values statement that nothing enforces. Governance without ethics produces a compliance process that satisfies its own paperwork while missing the harm entirely. Both failure modes are common, and organisations frequently have one while believing they have both.

IN PRACTICE

A mid-sized firm discovers it is already running four AI tools nobody formally approved: a CV screening plug-in, a customer chatbot, a scheduling assistant and a fraud-flagging feature inside its payments provider. Its first governance step is not a policy document. It is a one-page inventory listing what each tool does, who owns it, what data it touches and what happens if it is wrong. That single exercise almost always reveals which system needs attention first — and it is rarely the one people expected.

1.4 Why This Became Urgent

Three pressures arrived at roughly the same time, and any one of them would have been enough on its own.

The first is legal. UK data protection law already applies in full to AI systems that process personal data, and it has done for years — this is not new law waiting to be written. The Information Commissioner’s Office has issued specific guidance on AI and enforced against AI-driven processing. Other regulators have followed within their own remits.

The second is scale. A single biased hiring manager affects the candidates they personally see. A biased screening model affects every candidate, consistently, and produces no paper trail that looks like discrimination. Automation does not create new kinds of unfairness so much as it industrialises the existing kinds and makes them harder to see.

The third is reputational. AI failures are unusually legible to the public. They produce clear victims, screenshots, and a story that requires no technical knowledge to understand. The reputational damage typically arrives before any regulatory finding, and it is frequently the more expensive of the two.

CASE STUDY — OFQUAL AND THE 2020 A-LEVEL GRADES

In August 2020, with exams cancelled by COVID-19, England’s exams regulator Ofqual used a statistical model to assign A-level grades. The model combined teacher assessments with each school’s historic results. Roughly 40% of teacher-predicted grades were downgraded.

The effect was not evenly distributed. Because the model leaned on historic school performance, high-achieving pupils at large state schools with mixed results were pulled down, while pupils at small — often independent — schools were more likely to be marked up. The model was doing exactly what it was designed to do; the design encoded the assumption that a school’s past predicts its pupils’ present.

Within days the Government reversed the decision, and Ofqual’s chief regulator resigned. It remains the clearest British example of a system failing simultaneously on fairness, on transparency, and on accountability — and of what happens when the people affected cannot see or challenge the reasoning. Module 4 returns to it in detail.

KEY TAKEAWAYS

- AI ethics sets the standard; AI governance makes an organisation meet it consistently and prove that it did. Neither substitutes for the other.
- The practical definition of AI is any system deriving outputs from patterns in data rather than from explicitly written rules — regardless of what the vendor calls it.
- Most organisations already use AI they have not inventoried; the inventory, not the policy, is the correct first step.
- The three pressures driving this are legal obligation, the scale at which automation operates, and the unusual public legibility of AI failure.
- UK law already applies to AI in full. Nothing in this course depends on legislation that has not yet been passed.

FURTHER READING

- Information Commissioner’s Office — Guidance on AI and data protection, ico.org.uk
- DSIT — A pro-innovation approach to AI regulation (white paper), gov.uk
- OECD.AI Policy Observatory — oecd.ai

CHECK-IN QUIZ

1. A department head states that her team does not use AI. Her team uses a supplier tool that ranks sales leads by likelihood to convert, trained on the supplier's historic data. The correct conclusion is that:

- A. She is right, because the team did not build or configure the model
- B. The tool only counts as AI if it processes personal data
- C. The team uses AI and it belongs on the inventory
- D. The supplier is responsible for inventorying it

2. Your chief executive asks for an AI policy by Friday. The most useful thing you can realistically produce in that time is:

- A. A one-page inventory of the AI systems in use, with an owner named against each
- B. A values statement covering fairness, transparency and accountability
- C. A summary of the EU AI Act's risk tiers
- D. A staff training slide deck

3. An organisation has a well-drafted AI ethics policy, and no record of which systems exist, who owns them or when they were last reviewed. This is an example of:

- A. Governance without ethics
- B. Adequate accountability under UK GDPR
- C. A mature governance programme
- D. Ethics without governance

4. In the Ofqual case, the design assumption that produced the unfair outcomes was that:

- A. Teachers systematically over-predict their pupils' grades
- B. A school's historic results should shape the grades of its individual pupils
- C. Statistical models are more objective than human judgement
- D. Grades should be distributed normally across the population

5. A system is given permission to place orders with suppliers rather than recommend them to a buyer. In governance terms the significant change is that:

- A. An error becomes an event in the world rather than a proposal a human can decline
- B. It now requires more computing power
- C. It falls outside the scope of UK GDPR
- D. It no longer needs a named owner